

Kelompok 6

PRESENTASI FINAL PROJECT

Deployment Intelligence & Policy-as-Code

KELOMPOK 6



Aryasatya Alauddin

5027231082



Fiorenza Adelia Nalle

5027231053



Harwinda

5027231079



Fikri Aulia As Sa'adi

5027231026



**Muhammad Andrian Rizq
Prasetio**

5027231052

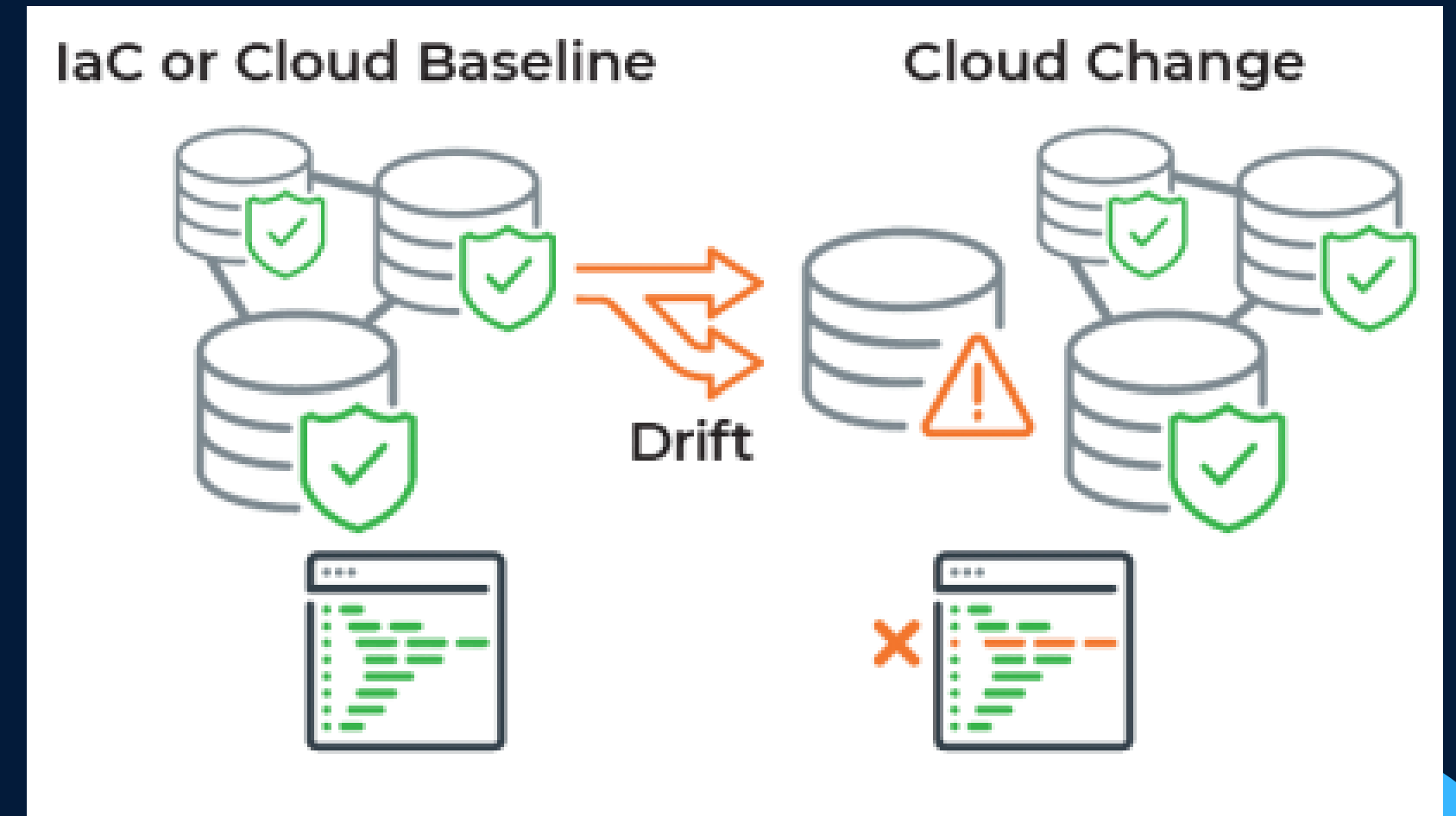


Nicholas Arya

50272310XX

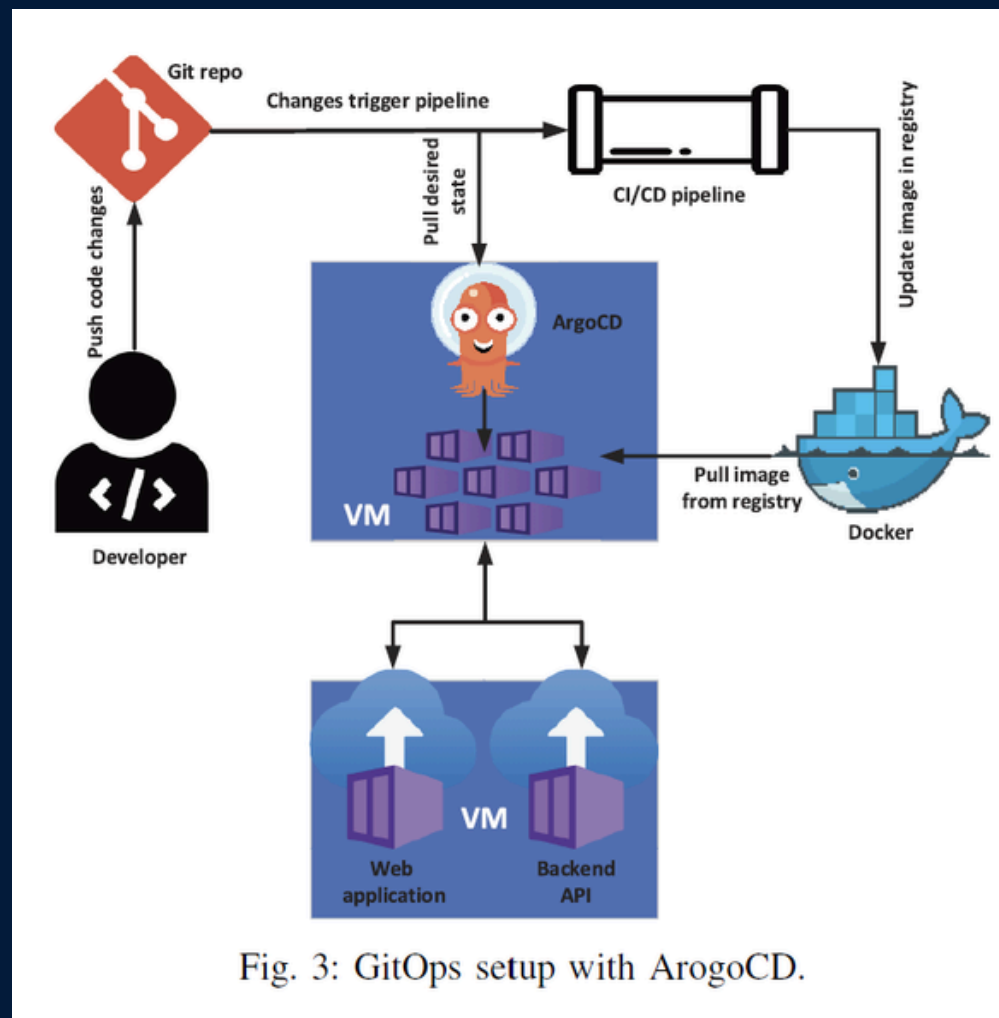
MASALAH PADA SISTEM CI/CD KLASIK (PUSH-BASED)

1. **Configuration Drift:** Jika server diretas/dihapus manual, sistem tidak sadar dan mati (MTTR lambat).
2. **Security Loophole:** Jenkins punya akses root/SSH ke dalam VPS, sangat berbahaya jika Jenkins diretas.
3. **Runtime Threat:** Tidak ada filter di dalam kluster. Image malware bisa masuk tanpa hambatan.



FONDASI LITERATUR ILMIAH

1. Configuration Management in Kubernetes Environments: A GitOps Approach



Membuktikan GitOps Pull-Based via ArgoCD menangani Configuration Drift secara real-time.

2. DevOps and Secure Cloud-Native Architectures for Finance

DevOps and Secure Cloud-Native Architectures for Finance

Sourabh Sanghi
Jawaharlal Nehru Technical University
Kukatpally, Hyderabad, India
sourabh.sanghi@gmail.com

Sunil Sudhakaran
Mahatma Gandhi University, Kottayam,
Kerala, India
sunilsudhakaran87@gmail.com

Vamsi Krishna Koganti
University of Missouri, Kansas City,
United States
vamcrulz09@gmail.com

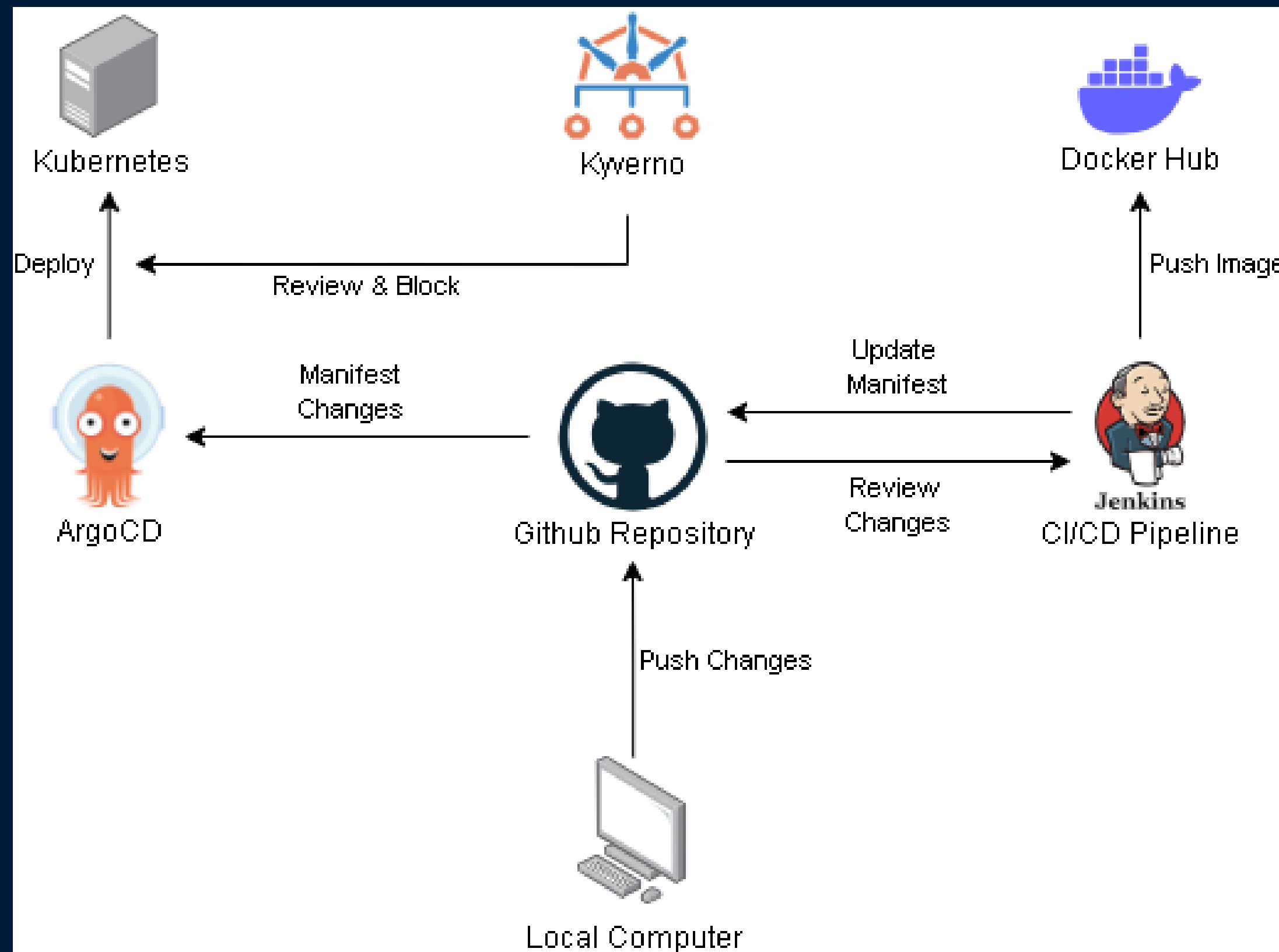
Narendra Ryali
Symbiosis Centre for Management Studies,
Nagpur,
Symbiosis International (Deemed
University), Pune, India
ryalinarendra@gmail.com

Abstract: With the ever-evolving and changing nature of financial services in the era of digital transformation, the concept of operational resilience, regulatory compliance, and innovation is highly important, which is achieved through secure cloud-native architectures and agility. This guide offers an end-to-end schema to build systems in the perspective of a financial institution - using DevSecOps, policy-as-code, Zero Trust, and infrastructure-as-code frameworks to build scalable, secure, and compliant systems. The revisit of ten historic research contributions, as well as the offering of a multi-layered model, fulfills the distinctive requirements and objectives of the financial IT-data sovereignty, rapid deployment, and high

recent technologies like fintech applications, blockchain services, and AI analysis in the financial sector (DevSecOps). These systems are supposed to be highly agile, scalable and secure, since financial information also has the elements of sensitivity and there are extremely stringent compliance requirements like (PCI-DSS, GDPR, SOX) which regulate financial transactions [2]. A paradigm shift in the development and deployment of the financial services under this paradigm is cloud-native architecture (microservices, containers and IaC). As devops and secure cloud-native techniques are put into practice, the amount of innovations

Mendemonstrasikan Triad SECURE. Validasi keamanan harus terjadi di runtime (Admission Controller) bukan hanya di CI

THE SOLUTION



Talk is cheap.
Show me the code.

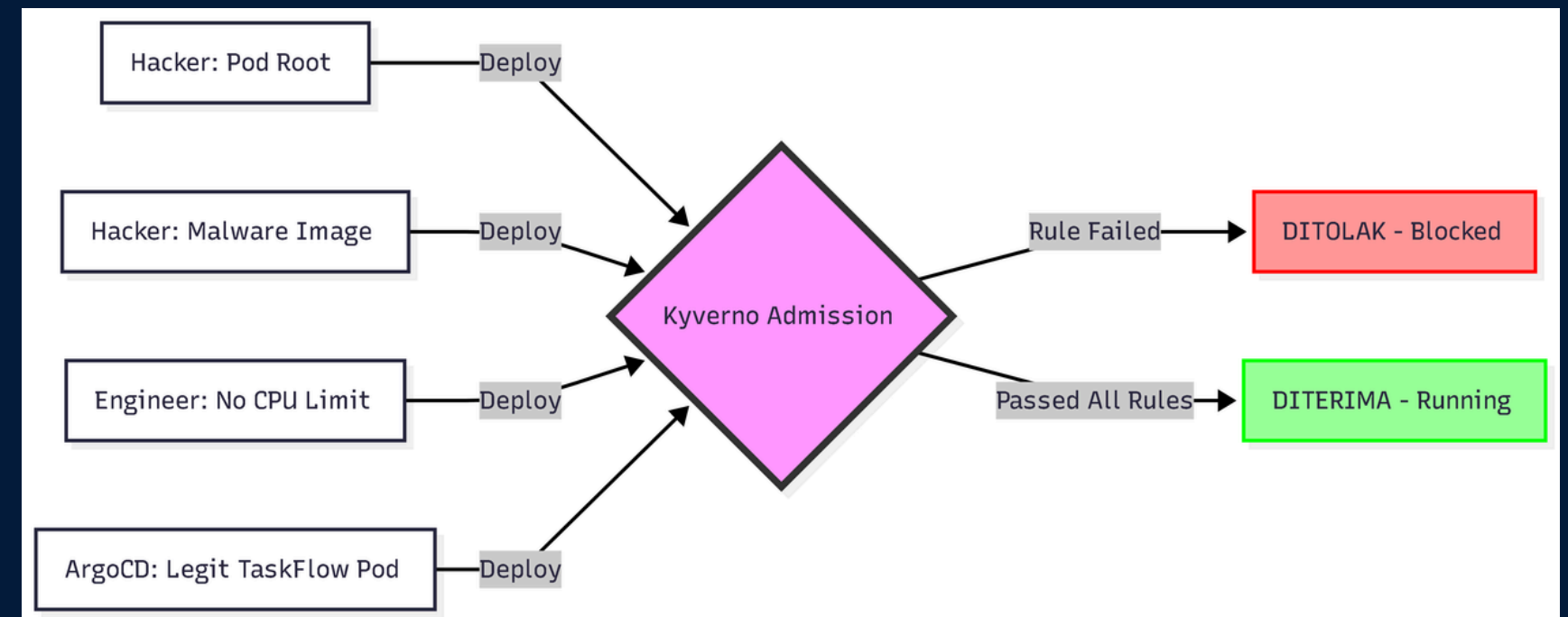
Linus Torvalds

EVALUASI



Peningkatan MTTR (Configuration Drift)

1. **Baseline (Sistem Lama):** 31.581 Detik (Besar human error / delay).
2. **Pasca-Enhancement (ArgoCD):** 2.038 Detik (100% otonom).
3. **Peningkatan Efisiensi:** >1500%.



Peningkatan MTTR (Configuration Drift)

1. Mencegah Escalation Privilege (Root User diblokir).
2. Mencegah Supply Chain Attack (Image asing diblokir).
3. Mencegah Resource Exhaustion (Wajib limitasi memori).

REFLEKSI DAN KESIMPULAN

1. **Resolusi Configuration Drift:** Transformasi ke Pull-Based GitOps terbukti sukses memangkas waktu pemulihan (MTTR) secara ekstrem dari 31.5 detik menjadi 2 detik (Efisiensi >1500%).
2. **Perlindungan Runtime Mutlak:** Implementasi Policy-as-Code (Kyverno) menutup celah Zero-Trust dengan keberhasilan 100% memblokir eksekusi manifes ilegal (Root User, No Limits, Untrusted Image).
3. **Kejutan Empiris (Refleksi):** Kecepatan Auto-Healing kluster saat merespons insiden di lapangan ternyata terjadi secara real-time, jauh melampaui asumsi teoretis awal kelompok kami.
4. **Pergeseran Paradigma (Refleksi):** Tantangan terbesar bukanlah soal teknis tools, melainkan mengubah mindset lama untuk berani mencabut hak akses deploy Jenkins dari dalam server demi keamanan.
5. **Rencana Pengembangan (Roadmap):** Jika diimplementasikan pada skala Enterprise, proyek ini harus dilanjutkan dengan penerapan Istio Service Mesh (mTLS) dan integrasi Trivy Admission Controller untuk perlindungan tingkat lanjut.



Kelompok 6

**THANKS FOR YOUR
ATTENTION!**